

DeepSeek-V2

A Strong, Economical, and Efficient
MoE Language Model

Pretrained on 8.1T tokens • Multi-source corpus • Extended Chinese data

arxiv.org/abs/2405.04434

236B

Total Parameters

21B

Activated / Token

128K

Context Length

The LLM Scaling Dilemma: Performance vs. Cost & Speed

Scaling = Better Intelligence

Larger models consistently yield higher accuracy across benchmarks

Training Cost Explosion

Compute requirements scale superlinearly with parameter count

KV Cache Bottleneck

Multi-Head Attention stores full K/V per head per token; memory grows linearly with sequence length during generation

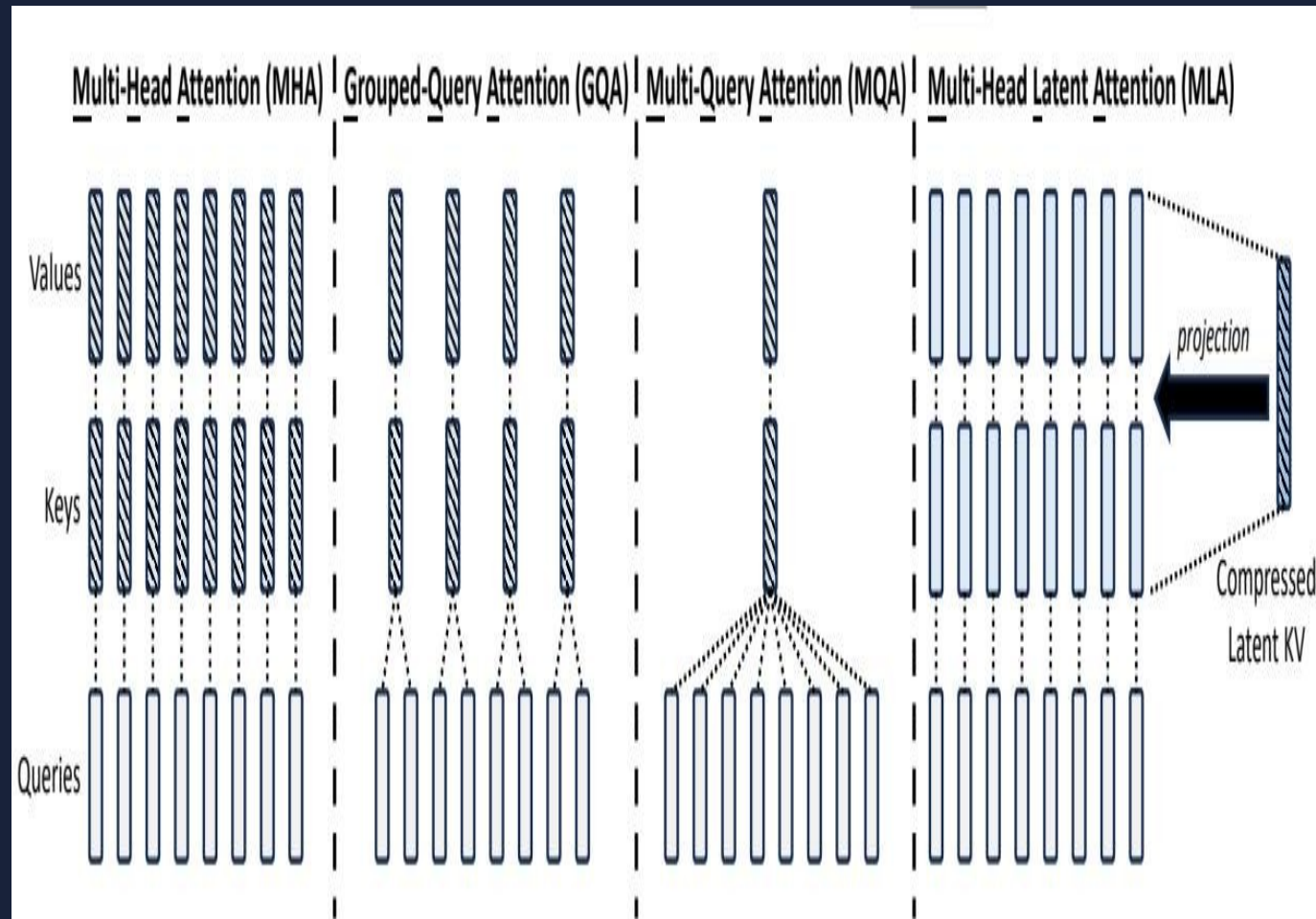
Inference Throughput Wall

Dense models underutilize hardware—every token activates all parameters

The Opportunity

- ▶ MoE: Activate a fraction of parameters per token → decouple model capacity from compute
- ▶ Efficient Attention: Compress KV cache without sacrificing quality
- ▶ Together: Train a 236B model cheaper than a 67B dense model, with dramatically faster inference

Multi-Head Latent Attention Compresses KV Cache by 93.3%



Low-Rank Compression

Full K and V vectors compressed into a compact latent vector c_t^{KV}

On-the-Fly Reconstruction

Keys, values, and positional keys reconstructed via learned projections

Decoupled RoPE

Positional encoding applied separately to avoid interfering with compression

Result

Better performance than MHA
93.3% less KV cache than standard attention

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Shared Experts (N_s)

- ▶ Always activated for every token
- ▶ Capture common cross-domain knowledge
- ▶ Reduce redundancy among routed experts

Routed Experts (N_r)

- ▶ Sparsely activated via top- K_r routing
- ▶ Fine-grained segmentation increases specialization
- ▶ Each token routed to experts on $\leq M$ devices

Device-Limited Routing

Constrains each token to experts on at most M devices, minimizing cross-node communication overhead

Auxiliary Load Balancing

Loss functions enforce balanced expert utilization across both experts and devices

Token-Dropping Strategy

Prevents expert overflow during training; no tokens dropped during inference

42.5% Training Cost Reduction & 5.76× Generation Throughput vs. DeepSeek 67B

DeepSeek 67B (Dense Baseline)

Architecture	Dense Transformer
Parameters	67B (all activated)
Training Cost	1.0× (reference)
KV Cache	1.0× (reference)
Max Gen Throughput	1.0× (reference)
MMLU	73.0

DeepSeek-V2 (MoE)

Architecture	MLA + DeepSeekMoE
Parameters	236B total / 21B activated
Training Cost	42.5% lower
KV Cache	93.3% smaller
Max Gen Throughput	5.76× higher
MMLU	78.5 (+5.5)

More capable, dramatically cheaper to train, and nearly 6× faster at inference

DeepSeek-V2 Tops MMLU at Only 21B Activated Parameters



DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

Alignment Pipeline

1. Supervised Fine-Tuning (SFT)
1.5M conversational sessions
2. Reinforcement Learning
Group Relative Policy Optimization (GRPO)

Chinese Language Leadership

- ▶ Outperforms all open-source models on AlignBench
- ▶ Surpasses most closed-source models in Chinese
- ▶ Demonstrates exceptional multilingual alignment

AlpacaEval 2.0

38.9%

Length-controlled
win rate

MT-Bench

8.97

Overall score
(English)

AlignBench

7.91

Overall score
(Chinese)

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

MLA: Superior KV-Cache Solution

Achieves better performance than MHA while reducing KV cache by 93.3% — strictly dominates GQA and MQA for inference efficiency.

DeepSeekMoE: Economical Training at Scale

Fine-grained experts with shared expert isolation enable training a 236B model at 42.5% lower cost than a 67B dense model.

Top-Tier Performance, Fraction of Compute

Only 21B activated parameters needed to match or exceed 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) on MMLU.

Production-Ready Throughput

5.76× generation throughput improvement over DeepSeek 67B makes large-scale deployment practical and cost-effective.

Strongest Open-Source Chinese Chat Model

DeepSeek-V2 Chat (RL) outperforms all open-source and most closed-source models on Chinese alignment benchmarks.

DeepSeek-V2: Pareto-optimal in the performance-efficiency frontier